

K-theory of $\mathbb{C}P^n$

Theorem 0.1. Let $L_1, \dots, L_n$ be line bundles over X and let H be the standard bundle $(\mathcal{O}(-1))$ over $P(L_1 \oplus \dots \oplus L_n)$. Then, the map

$$\frac{K(X)(t)}{\prod_{i=1}^n (t - [L_i^*])} \xrightarrow{\sim} K(P(L_1 \oplus \dots \oplus L_n))$$

$$t \mapsto [H]$$

induces an isomorphism of $K(X)$ -modules.

Proof. Let $p : P(E) \rightarrow X$ be a vector bundle. If L is a line-bundle over X . Then we have an isomorphism between

$$p(E) \xrightarrow{\sim} p(E \otimes L)$$

$$v \mapsto v \otimes l$$

for some $l \in L$. This map takes the standard hyperplane bundle H over $P(E)$ to the bundle $H \otimes p^*L$ over $p(E \otimes L)$.

Multiplying each bundle L_i by L_1^{-1} replaces $[H]$ by $[H][L_1]$ and gives an isomorphism between $P(L_1 \oplus \dots \oplus L_n) \simeq P(1 \oplus \dots \oplus P_n)$. So WLOG, we can assume that $L_1 = 1$.

We will argue by induction on n . The case $n = 0$ being trivial. For brevity write $P_m = P(L_1 \oplus \dots \oplus L_m)$ for $1 \leq m \leq n$. We have inclusions $X \cong P_1 \rightarrow P_2 \rightarrow \dots \rightarrow P_n$. If H_m denotes the hyperplane bundle over P_m , then $H_m|_{P_{m-1}} \cong H_{m-1}$.

We have a commutative diagram

$$\begin{array}{ccccc} P_{n-1} & \xrightarrow{s} & P(H_{n-1}^* + 1) & \xleftarrow{\text{red}} & q^*(H_n) \cong G \\ \downarrow & & \downarrow q & & \downarrow \text{red} \\ X \approx P_1 & \longrightarrow & P_n & \xleftarrow{\text{red}} & H_n \end{array}$$

The map $q : P(H^* + 1) \rightarrow P_n$ is defined by sending $[v_1 : \dots : v_n]_{[w_1 : \dots : w_{n-1}]} \mapsto [v_1 : \dots : v_n]$ where $[v_1 : \dots : v_{n-1}]$ is a multiple (possibly 0) of $[w_1 : \dots : w_{n-1}]$ and s the zero section. This induces a homeomorphism

$$\frac{P^*(H_{n-1} \oplus 1)}{s(P_{n-1})} \cong \frac{P_n}{i(P_1)}$$

$q^*(H_n) = G$ is the standard hyperplane bundle over $P(H^* + 1)$.

From the Periodicity theorem we know that

$$KP(1 \oplus H_{n-1}^*) \approx \frac{KP_{n-1}[G]}{(G-1)(G-H_{n-1}^*)} \approx KP_{n-1} \oplus K(P(H_{n-1}^*), sP_{n-1})$$

Since $s^*[G] = [1]$ it follows from the long exact sequence of cohomology groups that $k(P(H_{n-1}^* + 1), s(P_{n-1}))$ is the submodule generated by $[G] - 1$ and that on this submodule multiplication by $[G]$ and $[H_{n-1}]$ coincide.

Translating to the pair (P_n, P_1) , we get that $K(P_n, P_1)$ is a free KP_{n-1} module with generator $[H] - 1$ and

$$H_{n-1} \cdot x = [H_n] \cdot x \quad \forall x \in K(P_n, P_1)$$

We now use the inductive assumption that the proposition holds for $n - 1$ in place of n and thus may assume the isomorphism

$$KP_{n-1} = \frac{KX[H_{n-1}]}{\prod([L_i][H_{n-1}] - 1)}$$

We have a commutative diagram

$$\begin{array}{ccc} \frac{KX(t)}{\prod([L_i]t-1)} & \xrightarrow{t \mapsto H_{n-1}^*} & KP_{n-1} \\ t-1 \downarrow & & \downarrow H-1 \\ \frac{(t-1)KX(t)}{(t-1)\prod([L_i]t-1)} & \xrightarrow{t \mapsto H} & K(P_n, P_1) \end{array}$$

X is a retract of P_1 and we have a decomposition $KP_n \simeq KX \oplus K(P_n, X)$. Complementing the bottom row by the term KX on both sides gives the KX algebra isomorphism

$$\frac{KX[H]}{\prod(H - [L_i^{-1}])} \simeq KP_n$$

completing the induction step. □